

Dhairya Shah Lecture 15 Scribe



L15_L17.pdf
PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** constructed **strictly from the lecture slides you provided**.

All definitions, notation, assumptions, results, and worked examples are taken directly from the lecture slides in .

No external explanations or additional examples have been introduced.

Lecture Scribe

Joint Random Variables and Joint Distributions

(Based only on Lecture Slides L15–L17)

Source:

1. Motivation: Why Study Joint Random Variables

In many real-world problems, we are interested in **two (or more) random variables simultaneously**.

Examples given in the lecture slides include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples illustrate that **two variables may be related** and must be analyzed **jointly rather than separately**.

Example Applications

Example 1: Smart Transportation

Random variables:

- X = Vehicle Speed
- Y = Traffic Density

Question considered:

Can speed be high when traffic density is high?

This indicates that **speed and traffic density may be correlated random variables**.

Applications mentioned in slides:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Random variables:

- X = Signal-to-Noise Ratio (SNR)
- Y = Data Throughput

Reason for studying jointly:

Throughput depends on SNR, and both vary due to:

- fading
- interference
- congestion

Applications mentioned:

- 5G scheduling
 - adaptive modulation
 - network planning
-

Example 3–5 (Slides)

Additional examples listed:

- Weather prediction: X = rainfall, Y = temperature
 - Drone delivery: X = wind speed, Y = delivery time
 - Rolling two dice
-

Key Idea

When two random variables occur together:

- They have a **joint distribution**
- This allows computation of probabilities involving both variables
- It helps analyze **relationships between variables**

If variables are not independent, the dependence can be measured using:

- covariance
 - correlation
-

2. Discrete Case: Joint PMF

Setup and Notation

Suppose X and Y are **discrete random variables**.

Let

$$X = \{x_1, x_2, \dots, x_n\}$$

$$Y = \{y_1, y_2, \dots, y_m\}$$

Then the ordered pair (X, Y) takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

Definition (Joint PMF)

The **joint probability mass function** is defined as

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that:

- $X = x_i$
- and $Y = y_j$

occur **simultaneously**.

Joint PMF Table Representation

The joint PMF can be organized in a **table format**.

Rows correspond to values of X and columns correspond to values of Y .

Example structure from the slide:

$X \backslash Y$	y_1	y_2	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_m)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_m)$
...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_m)$

Key Properties of Joint PMF

1. Probability bounds

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals 1

$$\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1$$

3. Worked Example: Rolling Two Dice

Experiment

Roll two fair dice.

Define:

X = value on the first die

T = total (sum) of both dice

Each outcome (i, j) occurs with probability:

$$\frac{1}{36}$$

because there are **36 equally likely outcomes**.

Joint Probability Table

A table of (X, T) is constructed where:

- Rows correspond to $X = 1, 2, \dots, 6$
- Columns correspond to $T = 2, 3, \dots, 12$

Entries are either:

- $1/36$ if the combination is possible
- 0 otherwise

Example entries from the slide table (page showing table):

$X \backslash T$	2	3	4	5	6	7
1	1/36	1/36	1/36	1/36	1/36	1/36

$X \backslash T$	2	3	4	5	6	7
2	0	1/36	1/36	1/36	1/36	1/36
3	0	0	1/36	1/36	1/36	1/36
4	0	0	0	1/36	1/36	1/36
5	0	0	0	0	1/36	1/36
6	0	0	0	0	0	1/36

Observations from Slide

- Entries are **either 0 or 1/36**
- X and T are **not independent**.

4. Continuous Case: Joint PDF

The continuous case is analogous to the discrete case.

Relationship described in slides:

Discrete Case	Continuous Case
Discrete values	Continuous intervals
Joint PMF	Joint PDF
Sums	Double integrals

Domain of Random Variables

If

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) lies in the region:

$$[a, b] \times [c, d]$$

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** if:

$$P((X, Y) \text{ lies in a small rectangle } dx dy) \approx f(x, y) dx dy$$

Key Properties

1.

$$f(x, y) \geq 0$$

2.

$$\int_c^d \int_a^b f(x, y) dx dy = 1$$

Important note from slide:

The joint PDF $f(x, y)$ **may take values greater than 1**, because it is a **density**, not a probability.

5. Worked Example (Continuous Joint PDF)

Given

$$f(x, y) = 4xy$$

for

$$0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus the pair (X, Y) lies in the **unit square**

$$[0, 1]^2$$

Tasks given in the slides:

1. Show $f(x, y)$ is a valid joint PDF
2. Visualize the event

$$A = \{X < 0.5, Y > 0.5\}$$

3. Compute $P(A)$

Step 1: Check Validity (Normalization)

Compute the total probability:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x :

$$\begin{aligned} & \int_0^1 4xy \, dx \\ &= 4y \int_0^1 x \, dx \\ &= 4y \left[\frac{x^2}{2} \right]_0^1 \\ &= 4y \cdot \frac{1}{2} \\ &= 2y \end{aligned}$$

Now integrate with respect to y :

$$\begin{aligned} & \int_0^1 2y \, dy \\ &= [y^2]_0^1 \\ &= 1 \end{aligned}$$

Thus:

- $f(x, y) \geq 0$
- total probability = 1

Therefore $f(x, y)$ is a valid joint PDF.

Step 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

First integrate with respect to y :

$$\begin{aligned} & \int_{0.5}^1 4xy \, dy \\ &= 4x \int_{0.5}^1 y \, dy \\ &= 4x \left[\frac{y^2}{2} \right]_{0.5}^1 \\ &= 4x \left(\frac{1}{2} - \frac{0.25}{2} \right) \\ &= 4x(0.375) \\ &= 1.5x \end{aligned}$$

Now integrate with respect to x :

$$\begin{aligned} & \int_0^{0.5} 1.5x \, dx \\ &= 1.5 \left[\frac{x^2}{2} \right]_0^{0.5} \\ &= 1.5 \cdot \frac{0.25}{2} \\ &= \frac{3}{16} \end{aligned}$$

Thus:

$$P(A) = \frac{3}{16}$$

6. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation:

$$X \leq x, Y \leq y$$

means the event

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

The **joint cumulative distribution function** is defined as

$$F(x, y) = P(X \leq x, Y \leq y)$$

7. Joint CDF: Continuous Case

If X and Y have joint density $f(x, y)$ over the region $[a, b] \times [c, d]$, then

$$F(x, y) = \int_c^y \int_a^x f(u, v) du dv$$

Recovering Joint PDF from CDF

Since two variables are involved, **partial derivatives** are used:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

8. Joint CDF: Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation from the slide:

- Add probabilities in the table
- Include rows where $x_i \leq x$
- Include columns where $y_j \leq y$

9. Properties of the Joint CDF

The joint CDF $F(x, y)$ must satisfy:

Property 1

$F(x, y)$ is **non-decreasing**.

If x or y increases, $F(x, y)$ must stay constant or increase.

Property 2

At the **lower-left corner** of the joint range:

$$F(x, y) = 0$$

If the lower-left corner is $(-\infty, -\infty)$:

$$\lim_{(x, y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$$

Property 3

At the **upper-right corner** of the joint range:

$$F(x, y) = 1$$

If the upper-right corner is (∞, ∞) :

$$\lim_{(x, y) \rightarrow (\infty, \infty)} F(x, y) = 1$$

✓ This lecture scribe contains only material derived from the slides in , structured for **exam revision** with:

- Definitions and notation
 - Assumptions and conditions
 - Theorems and properties
 - Step-by-step worked example
 - Intermediate derivation steps
-

If you want, I can also generate **Version 2 of the same lecture scribe (same slides but different structure)** so you can compare outputs and choose the best one for exam revision.



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All definitions, notation, conditions, statements, and worked examples come directly from the slides in .

No external explanations, intuition, or additional examples are included.

Lecture Scribe

Joint Random Variables and Joint Distributions

Source:

1. Motivation: Why Study Joint Random Variables

In science and real-life situations, we are often interested in **two or more random variables at the same time**. (Slide page 3)

Examples listed in the lecture slides:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These cases require analyzing **two variables simultaneously**.

Application Examples from Slides

Example 1: Smart Transportation

Random variables:

X = Vehicle Speed

Y = Traffic Density

Question posed in slides:

Can speed be high when traffic density is also high?

Conclusion mentioned:

X and Y are **correlated random variables**.

Applications mentioned:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

Random variables:

X = Signal-to-Noise Ratio (SNR)

Y = Data Throughput

Reason to study jointly:

Throughput depends on SNR, and both vary due to:

- fading
- interference
- congestion

Applications:

- 5G scheduling
 - adaptive modulation
 - network planning
-

Additional Examples from Slides

- Weather prediction: X = rainfall, Y = temperature
 - Drone delivery: X = wind speed, Y = delivery time
 - Rolling two dice (joint PMF example)
-

Key Idea from Slides

Random variables in such situations have a **joint distribution** that allows computation of probabilities involving both variables and helps understand the **relationship between them**.

Special cases mentioned:

- If variables are **independent**, analysis is simpler.
- If not independent, dependence may be measured using:
 - covariance
 - correlation (Slide page 6).

2. Discrete Case: Joint Probability Mass Function (Joint PMF)

Assumptions and Setup

Suppose:

$$X, Y$$

are **discrete random variables**.

Let the possible values be:

$$X = \{x_1, x_2, \dots, x_n\}$$

$$Y = \{y_1, y_2, \dots, y_m\}$$

Then the ordered pair:

$$(X, Y)$$

takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

(Slide page 7).

Definition: Joint PMF

The **joint probability mass function** is defined as:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that:

- $X = x_i$
- $Y = y_j$

occur **simultaneously**. (Slide page 8).

3. Joint PMF Table Representation

The joint PMF is organized in a **table format** (Slide page 10).

General structure:

$X \setminus Y$	y_1	y_2	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_m)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_m)$
...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_m)$

Key Properties of Joint PMF

1. Probability bounds:

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals 1:

$$\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1$$

(Slide page 10).

4. Worked Example: Rolling Two Dice

Experiment

Roll **two fair dice**. (Slide page 11)

Define:

X = value on the first die

T = total (sum) of both dice

Each outcome (i, j) occurs with probability:

$$\frac{1}{36}$$

because there are 36 equally likely outcomes (Slide page 12).

Construct Joint Probability Table of (X, T)

Rows represent values of $X = 1, 2, 3, 4, 5, 6$

Columns represent values of $T = 2, 3, \dots, 12$

Entries are either:

- $1/36$ if the pair (X, T) is possible
- 0 otherwise

Example rows from the table shown in the slides (page 14):

$X \backslash T$	2	3	4	5	6	7
1	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
2	0	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
3	0	0	$1/36$	$1/36$	$1/36$	$1/36$

Observations from the Slides

- Entries are **either 0 or $1/36$**
 - X and T are **not independent**. (Slide page 14).
-

5. Continuous Case: Joint PDF

The continuous case is analogous to the discrete case.

Relationship stated in the slides:

Discrete Case	Continuous Case
Discrete values	Continuous intervals
Joint PMF	Joint PDF

(Slide page 15).

Domain of Continuous Variables

If:

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) takes values in the region:

$$[a, b] \times [c, d]$$

(Slide page 16).

Definition: Joint Probability Density Function

A function $f(x, y)$ is called the **joint PDF** of X and Y if:

$$P((X, Y) \text{ lies in a small rectangle of area } dx dy) \approx f(x, y) dx dy$$

(Slide page 17).

Key Properties of Joint PDF

1.

$$f(x, y) \geq 0$$

2.

$$\int_c^d \int_a^b f(x, y) dx dy = 1$$

Important note from slides:

The joint PDF **may take values greater than 1**, because it is a **density and not a probability** (Slide page 20).

6. Continuous Example (Worked Example)

Given

$$f(x, y) = 4xy$$

with domain:

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

Thus:

$$(X, Y) \in [0, 1]^2$$

(Slide page 21).

Tasks in Slides

1. Show $f(x, y)$ is a valid joint PDF
2. Visualize event

$$A = \{X < 0.5, Y > 0.5\}$$

3. Compute $P(A)$
-

Step 1: Check Validity (Normalization)

Compute:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x :

$$\int_0^1 4xy \, dx$$

Factor constant:

$$4y \int_0^1 x \, dx$$

Compute integral:

$$\int_0^1 x \, dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$$

Therefore:

$$4y \times \frac{1}{2} = 2y$$

Now integrate with respect to y :

$$\begin{aligned} \int_0^1 2y \, dy \\ &= [y^2]_0^1 \\ &= 1 \end{aligned}$$

Thus:

- $f(x, y) \geq 0$
- Total probability = 1

Hence $f(x, y)$ is a **valid joint PDF** (Slide page 25).

Step 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

First integrate with respect to y :

$$\begin{aligned} \int_{0.5}^1 4xy \, dy \\ &= 4x \int_{0.5}^1 y \, dy \end{aligned}$$

$$\begin{aligned}
&= 4x \left[\frac{y^2}{2} \right]_{0.5}^1 \\
&= 4x \left(\frac{1}{2} - \frac{0.25}{2} \right) \\
&= 4x(0.375) \\
&= 1.5x
\end{aligned}$$

Now integrate with respect to x :

$$\begin{aligned}
&\int_0^{0.5} 1.5x \, dx \\
&= 1.5 \left[\frac{x^2}{2} \right]_0^{0.5} \\
&= 1.5 \times \frac{0.25}{2} \\
&= \frac{3}{16}
\end{aligned}$$

Thus:

$$P(A) = \frac{3}{16}$$

(Slide page 26).

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation used in slides:

$$X \leq x, \quad Y \leq y$$

means the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

(Slide page 28).

Definition

The **joint cumulative distribution function** is defined as:

$$F(x, y) = P(X \leq x, Y \leq y)$$

(Slide page 29).

8. Joint CDF for Continuous Variables

If X and Y have joint density $f(x, y)$, then:

$$F(x, y) = \int_c^y \int_a^x f(u, v) du dv$$

(Slide page 32).

Recovering the Joint PDF

Because two variables are involved:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

(Slide page 33).

9. Joint CDF for Discrete Variables

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

(Slide page 36).

Interpretation from Slides

To compute $F(x, y)$:

1. Add probabilities in the table
2. Include rows where $x_i \leq x$

3. Include columns where $y_j \leq y$

(Slide page 37).

10. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies the following properties:

Property 1: Monotonicity

$F(x, y)$ is non-decreasing

If x or y increases, then $F(x, y)$ must **stay constant or increase**.

Property 2: Lower-Left Limit

$$F(x, y) = 0$$

at the **lower-left** of the joint range.

If the corner is $(-\infty, -\infty)$:

$$\lim_{(x,y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$$

Property 3: Upper-Right Limit

$$F(x, y) = 1$$

at the **upper-right** of the joint range.

If the corner is (∞, ∞) :

$$\lim_{(x,y) \rightarrow (\infty, \infty)} F(x, y) = 1$$

(Slide page 41).

✓ This lecture scribe strictly includes material from the provided slides and contains:

- Definitions and notation

- Assumptions and conditions
- Statements of results
- Step-by-step derivations shown in the slides
- Complete worked example with intermediate steps

All content is derived from for **exam revision purposes**.



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Lecture Scribe

Joint Random Variables and Joint Distributions

(CSE400 – Fundamentals of Probability in Computing)

Source:

1. Motivation: Why Study Joint Random Variables

In science and real-life situations we are often interested in **two (or more) random variables simultaneously**. (Slide page 3)

Examples given in the slides include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These situations require analyzing **multiple random variables together** rather than individually.

Example Applications

Example 1: Smart Transportation (Slide page 4)

Two random variables:

X = Vehicle Speed

Y = Traffic Density

Question posed in the slide:

Can speed be high when traffic density is also high?

Conclusion stated in slides:

X and Y are **correlated random variables**.

Applications mentioned:

- Adaptive traffic signals
 - Congestion prediction
 - Autonomous vehicle routing
-

Example 2: Wireless Network Performance

Random variables:

X = Signal-to-Noise Ratio (SNR)

Y = Data Throughput

Reason to study jointly:

Throughput depends on SNR and both vary due to:

- fading
- interference
- congestion

Applications:

- 5G scheduling
 - adaptive modulation
 - network planning
-

Additional Examples Mentioned

- Weather prediction: X = rainfall, Y = temperature
 - Drone delivery: X = wind speed, Y = delivery time
 - Rolling two dice (joint PMF example)
-

Key Concept from Slides

Random variables in such situations have a **joint distribution** which allows computation of probabilities involving both variables and helps understand the **relationship between variables**.

When variables are not independent, their dependence may be analyzed using:

- covariance
 - correlation (Slide page 6).
-

2. Discrete Case: Joint Probability Mass Function (Joint PMF)

Assumptions and Setup

Suppose:

$$X, Y$$

are **discrete random variables**.

Possible values:

$$X = \{x_1, x_2, \dots, x_n\}$$

$$Y = \{y_1, y_2, \dots, y_m\}$$

Then the ordered pair:

$$(X, Y)$$

takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \dots, (x_n, y_m)\}$$

(Slide page 7).

Definition (Joint PMF)

The **joint probability mass function** is defined as:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This gives the probability that:

- $X = x_i$
- $Y = y_j$

occur **simultaneously** (Slide page 8).

3. Joint PMF Table Representation

The joint PMF is often organized into a **table** (Slide page 10).

General structure:

$x \backslash y$	y_1	y_2	...	y_m
x_1	$p(x_1, y_1)$	$p(x_1, y_2)$	...	$p(x_1, y_m)$
x_2	$p(x_2, y_1)$	$p(x_2, y_2)$	...	$p(x_2, y_m)$
...	...	...	...	...
x_n	$p(x_n, y_1)$	$p(x_n, y_2)$	...	$p(x_n, y_m)$

Properties of Joint PMF

1.

$$0 \leq p(x_i, y_j) \leq 1$$

2.

$$\sum_{i=1}^n \sum_{j=1}^m p(x_i, y_j) = 1$$

(Slide page 10).

4. Worked Example: Rolling Two Dice

Experiment (Slide page 11)

Roll two fair dice.

Define:

X = value on the first die

T = total (sum) of both dice

Probability of Each Outcome

Each ordered pair (i, j) occurs with probability:

$$\frac{1}{36}$$

because there are 36 equally likely outcomes (Slide page 12).

Joint Probability Table for (X, T)

Rows represent values of X .

Columns represent values of T .

Entries are either:

- $1/36$
- 0

Example rows from the table shown on slide page 14:

$X \backslash T$	2	3	4	5	6	7
1	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
2	0	$1/36$	$1/36$	$1/36$	$1/36$	$1/36$
3	0	0	$1/36$	$1/36$	$1/36$	$1/36$

Observations (Slide page 14)

- Table entries are **either 0 or $1/36$** .
 - X and T are **not independent**.
-

5. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case is analogous to the discrete case.

Relationship stated in slides:

Discrete Case	Continuous Case
Discrete values	Continuous intervals
Joint PMF	Joint PDF
Sums	Double integrals

(Slide page 15).

Domain of Continuous Variables

If:

$$X \in [a, b]$$

$$Y \in [c, d]$$

then the pair (X, Y) lies in:

$$[a, b] \times [c, d]$$

(Slide page 16).

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** if:

$$P((X, Y) \text{ lies in a small rectangle of area } dx dy) \approx f(x, y) dx dy$$

(Slide page 17).

Properties of Joint PDF

1.

$$f(x, y) \geq 0$$

2.

$$\int_c^d \int_a^b f(x, y) dx dy = 1$$

Important note from slides:

The joint PDF **may take values greater than 1**, because it is a **density and not a probability** (Slide page 20).

6. Continuous Worked Example

Given (Slide page 21)

$$f(x, y) = 4xy$$

for:

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

Thus:

$$(X, Y) \in [0, 1]^2$$

Tasks

1. Show $f(x, y)$ is a valid joint PDF
2. Visualize the event

$$A = \{X < 0.5, Y > 0.5\}$$

3. Compute $P(A)$
-

Step 1: Check Validity (Normalization)

Compute:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Integrate with respect to X

$$\int_0^1 4xy \, dx$$

$$\begin{aligned}
 &= 4y \int_0^1 x \, dx \\
 &= 4y \left[\frac{x^2}{2} \right]_0^1 \\
 &= 4y \cdot \frac{1}{2} \\
 &= 2y
 \end{aligned}$$

Integrate with respect to y

$$\begin{aligned}
 &\int_0^1 2y \, dy \\
 &= [y^2]_0^1 \\
 &= 1
 \end{aligned}$$

Therefore:

- $f(x, y) \geq 0$
- total probability = 1

Hence $f(x, y)$ is a **valid joint PDF** (Slide page 25).

Step 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Probability:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

Integrate with respect to y

$$\begin{aligned}
 &\int_{0.5}^1 4xy \, dy \\
 &= 4x \int_{0.5}^1 y \, dy
 \end{aligned}$$

$$\begin{aligned}
&= 4x \left[\frac{y^2}{2} \right]_{0.5}^1 \\
&= 4x \left(\frac{1}{2} - \frac{0.25}{2} \right) \\
&= 4x(0.375) \\
&= 1.5x
\end{aligned}$$

Integrate with respect to x

$$\begin{aligned}
&\int_0^{0.5} 1.5x \, dx \\
&= 1.5 \left[\frac{x^2}{2} \right]_0^{0.5} \\
&= 1.5 \cdot \frac{0.25}{2} \\
&= \frac{3}{16}
\end{aligned}$$

Thus:

$$P(A) = \frac{3}{16}$$

(Slide page 26).

7. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

Notation used in slides:

$$X \leq x, \quad Y \leq y$$

means the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

(Slide page 28).

Definition

The **joint CDF** is defined as:

$$F(x, y) = P(X \leq x, Y \leq y)$$

(Slide page 29).

8. Joint CDF – Continuous Case

If X and Y have joint density $f(x, y)$, then:

$$F(x, y) = \int_c^y \int_a^x f(u, v) du dv$$

(Slide page 32).

Recovering Joint PDF

Because two variables are involved:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

(Slide page 33).

9. Joint CDF – Discrete Case

If X, Y are discrete random variables with joint PMF $p(x_i, y_j)$:

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

(Slide page 36).

Interpretation from Slides

To compute $F(x, y)$:

- Add probabilities in the table
- Include rows with $x_j \leq x$

- Include columns with $y_j \leq y$

(Slide page 37).

10. Properties of Joint CDF

The joint CDF $F(x, y)$ satisfies the following properties:

Property 1 (Monotonicity)

$F(x, y)$ is non-decreasing

If x or y increases, $F(x, y)$ stays constant or increases.

Property 2 (Lower-left corner)

$$F(x, y) = 0$$

If the corner is $(-\infty, -\infty)$:

$$\lim_{(x, y) \rightarrow (-\infty, -\infty)} F(x, y) = 0$$

Property 3 (Upper-right corner)

$$F(x, y) = 1$$

If the corner is (∞, ∞) :

$$\lim_{(x, y) \rightarrow (\infty, \infty)} F(x, y) = 1$$

(Slide page 41).

✓ This lecture scribe includes:

- Definitions and notation
- Assumptions and conditions
- Statements of results
- Step-by-step derivations

- Worked examples with intermediate steps

All material is taken **strictly from the lecture slides** in for exam revision purposes.